

MANIPAL UNIVERSITY JAIPUR
Department of Artificial Intelligence and Machine Learning
Jaipur, Rajasthan, India

ASSIGNMENT No. 1

Programme	B.Tech. Artificial Intelligence & Machine Learning	Semester / Section	5 th Sem (All Section)
Course Title	Automata and Formal Languages	Course Code	 (as per Course Handout)
Academic Session		Assignment No.	01
Course Coordinator	Dr. Shinnu Jangra, Assistant Professor	Maximum Marks	20 (scaled to 10 as per CWS)
Date of Issue		Last Date of Submission	28 September 2026
Unit / Modules Covered	Regular Languages, CFLs, Turing Machines, Decidability, Complexity	Mode of Submission	Handwritten hard copy (A4)

1. Instructions to Students

1. Attempt all questions. The assignment carries 20 marks and will be scaled down to the 10-mark Assignment component of the Class Work Sessional (CWS).
2. The assignment must be handwritten on A4 sheets (one side only) and stapled, with the cover sheet at Annexure-I as the first page.
3. A hard copy is to be submitted in class on the due date. Early submission is allowed; late submission is not accepted under any circumstances.
4. Show complete work. Formal definitions (5-tuples / 7-tuples), transition tables, derivations, instantaneous descriptions and proof steps carry marks; a final answer alone does not.
5. Draw all automata neatly — mark the start state with an incoming arrow and every final state with a double circle and label every transition.
6. Write the question number and its CO–BL tag exactly as given in the question table before every answer.
7. Submissions found to be copied, in part or in full, from a classmate or from an online source will be awarded zero marks as per the University's academic integrity policy.
8. Evaluated assignments will be returned in class and must be retained by the student until the end of the semester for OBE / accreditation records.

2. Course Outcomes Addressed

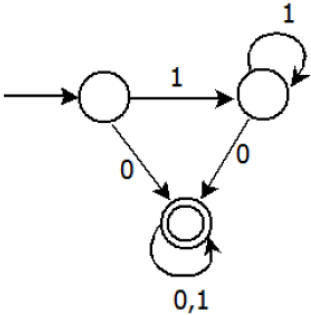
The CO statements below are derived from the course objectives stated in the course introduction; the numbering is to be verified against the approved Course Handout before circulation.

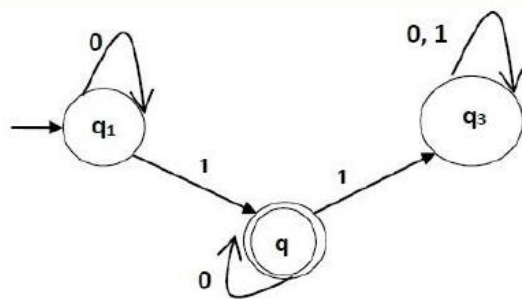
CO	Course Outcome — on completion of the course, the student will be able to:
CO1	Explain the fundamental concepts of automata theory and the theory of computation.
CO2	Identify the different formal language classes and the relationships among them.
CO3	Design grammar and recognizers (FA, RE, RG, PDA, TM) for different formal languages.

CO4	Prove or disprove propositions in automata theory using the properties of formal languages.
CO5	Determine the decidability and the intractability of computational problems.

3. Assignment Questions

Q. No.	Question	Marks	CO	BL
PART – A: Regular Languages (Finite Automata, Regular Grammars, Regular Expressions and their Properties) — 12 Marks				
1	(a) Let $\Sigma = \{0, 1\}$. Write the sets Σ^0, Σ^1 and Σ^2, and state the number of strings in Σ^5. Distinguish Σ^* from Σ^+ with one example each and define the length of a string and the concatenation of two strings. (b) Draw the Chomsky hierarchy as a containment diagram and complete a table giving, for each of the four classes, the type of grammar, the recognizing machine and one example language.	2 (1+1)	CO1	BL2
2	(a) Design a DFA over $\Sigma = \{0, 1\}$ that accepts every string ending with 01. Give the formal 5-tuple $(Q, \Sigma, \delta, q_0, F)$, the transition table and the state diagram, and trace the extended transition function (δ-hat) on the input 11001. (b) Design a DFA over $\Sigma = \{0, 1\}$ that accepts every string in which the number of 1's is divisible by 3. Write the language of your DFA in set-builder notation.	2 (1+1)	CO3	BL3
3	Consider the NFA $N = (\{q_0, q_1, q_2\}, \{0, 1\}, \delta, q_0, \{q_2\})$ with $\delta(q_0, 0) = \{q_0, q_1\}$, $\delta(q_0, 1) = \{q_0\}$, $\delta(q_1, 1) = \{q_2\}$ and $\delta(q_2, 0) = \delta(q_2, 1) = \emptyset$. (a) Convert N into an equivalent DFA by the subset construction using lazy creation. Show the transition table of the DFA and describe its language in English. (b) Apply the table-filling algorithm to the DFA obtained in (a), list the distinguishable state pairs and state whether the DFA is already minimal. Also state the worst-case number of DFA states obtained from an n-state NFA.	2 (1+1)	CO2	BL3
4	(a) Construct a right-linear grammar $G = (V, T, P, S)$ for $L = \{w \in \{a, b\}^* \mid w \text{ contains the substring } ab\}$ and show a leftmost derivation of the string aabb. (b) Convert the DFA designed in Q2(a) into an equivalent right-linear grammar (one variable per state; an ϵ-production for every final state). State the theorem that establishes the equivalence of regular grammars, finite automata and regular expressions.	2 (1+1)	CO3	BL3
5	(a) Write regular expressions over $\Sigma = \{0, 1\}$ for: (i) all strings that begin with 1 and end with 0; (ii) all strings containing at least two consecutive 0's; (iii) all strings of even length. (b) Construct an ϵ-NFA for the regular expression $(0 + 1)^*01$, compute the ϵ-closure of every state and convert the ϵ-NFA into an equivalent DFA. (c) State, with a proof or a counter-example, whether each identity is valid: (i) $(r + s)^* = r^* s^*$ (ii) $(r^*)^* = r^*$.	2 ($\frac{1}{2}+1+\frac{1}{2}$)	CO2	BL3

Q. No.	Question	Marks	CO	BL
6	(a) Using the closure of regular languages under intersection, prove that $L' = \{ w \in \{a, b\}^* \mid w \text{ has an equal number of a's and b's} \}$ is not regular. Also explain how emptiness and finiteness of a regular language are decided from its DFA.	2 (1+1)	CO4	BL4
7.	Find DFA for the following languages on $\Sigma = \{a, b\}$: (a) $L = \{w : w \bmod 3 = 0\}$ (b) $L = \{w : na(w) \bmod 3 > nb(w) \bmod 3\}$ (c) Consider the DFA given.  Which of the following are FALSE? (a) Complement of $L(A)$ is context-free. (b) $L(A) = L((11^*0+0)(0+1)^*0^*1^*)$. (c) For the language accepted by A, A is the minimal DFA. (d) A accepts all strings over $\{0, 1\}$ of length at least 2.	2 (1+1)	CO1	BL2
8.	Consider the set of strings on $\{0,1\}$. Design a DFA to accept all strings where every 00 is followed immediately by a 1.	2 (1+1)	CO1	BL2
9.	Construct a DFA equivalent to the NFA defined by the regular expression $(0+1)^*00$	2 (1+1)	CO2	BL3
10.	Represent the following sets by regular expression: (a) $\{1^{2n+1} \mid n > 0\}$ (b) {The set of all strings over $\{0,1\}$ which has at most two zeros.} (c) The set of all strings over $\{a, b\}$ beginning and ending with a.	2 (1+1)	CO2	BL3
11.	Construct a transition system corresponding to the following regular expressions: (a). $(ab + c^*)^*b$ (b). $a + bb + bab^*a$ (c). $(ab + a)^*(aa + b)$ (d). $(a^*b + b^*a)^*a$	2 (1+1)	CO2	BL3

Q. No.	Question	Marks	CO	BL																											
12.	Prove the following: a. Arden's theorem b. $(0^*1^*)^* = (0 + 1)^*$ c. $(ab)^* \neq (a^*b^*)$ d. $(r + s)^* \neq r^* + s^*$ e. $\lambda + 1^*(011)^*(1^*(011)^*)^* = (1 + 011)^*$	2 (1+1)	CO2	BL3																											
13.	Construct R.E. using Arden's theorem from the following FA's: a. <table border="1"><tr><td></td><td>a</td><td>B</td></tr><tr><td>$\rightarrow q1$</td><td>q1</td><td>q2</td></tr><tr><td>$*q2$</td><td>q3</td><td>q2</td></tr><tr><td>q3</td><td>$\varnothing$</td><td>$\varnothing$</td></tr></table> b. <table border="1"><tr><td></td><td>0</td><td>1</td></tr><tr><td>$\rightarrow^*$</td><td>q1</td><td>q2</td></tr><tr><td>q1</td><td></td><td></td></tr><tr><td>q2</td><td>q3</td><td>q2</td></tr><tr><td>q3</td><td>q1</td><td>q2</td></tr></table>		a	B	$\rightarrow q1$	q1	q2	$*q2$	q3	q2	q3	$\varnothing$	$\varnothing$		0	1	$\rightarrow^*$	q1	q2	q1			q2	q3	q2	q3	q1	q2	2 (1+1)	CO2	BL3
	a	B																													
$\rightarrow q1$	q1	q2																													
$*q2$	q3	q2																													
q3	$\varnothing$	$\varnothing$																													
	0	1																													
$\rightarrow^*$	q1	q2																													
q1																															
q2	q3	q2																													
q3	q1	q2																													
14.	Construct a regular expression corresponding to the automata given below – 	2 (1+1)	CO2	BL3																											
PART – B : Context-Free Languages, Turing Machines, Decidability and Complexity — 8 Marks																															
15	(a) Give context-free grammars for (i) $L_1 = \{ 0^n1^n \mid n \geq 0 \}$ and (ii) $L_2 =$ the set of balanced strings of parentheses. Draw the parse tree for the string 000111 using your grammar for L_1 . (b) Show that the grammar $E \rightarrow E + E \mid E \times E \mid (E) \mid a$ is ambiguous by drawing two distinct parse trees for $a + a \times a$, and rewrite it as an unambiguous grammar that enforces the usual precedence and associativity.	2 (1+1)	CO3	BL3																											
16.	Consider the following grammar- $\begin{aligned} S &\rightarrow ABC \\ A &\rightarrow a \\ B &\rightarrow b \\ C &\rightarrow c \end{aligned}$ Consider a string $w = abc$. Design all possible LMD and RMD.	2 (1+1)	CO3	BL3																											

Q. No.	Question	Marks	CO	BL
17	1. Consider the grammar- $S \rightarrow bB / aA$ $A \rightarrow b / bS / aAA$ $B \rightarrow a / aS / bBB$ For the string $w = bbaababa$, find- - Leftmost derivation - Rightmost derivation - Parse Tree	2 (1+1)	CO3	BL3
18.	Consider the following grammar and eliminate left recursion- $A \rightarrow ABd / Aa / a$ $B \rightarrow Be / b$ Consider the following grammar and eliminate left recursion- $E \rightarrow E + E / E \times E / a$ Consider the following grammar and eliminate left recursion- $E \rightarrow E + T / T$ $T \rightarrow T \times F / F$ $F \rightarrow id$	2 (1+1)	CO3	BL3
19.	Check whether the given grammar is ambiguous or not- $S \rightarrow SS$ $S \rightarrow a$ $S \rightarrow b$ Check whether the given grammar is ambiguous or not- $S \rightarrow A / B$ $A \rightarrow aAb / ab$ $B \rightarrow abB / \epsilon$ Check whether the given grammar is ambiguous or not- $S \rightarrow AB / C$ $A \rightarrow aAb / ab$ $B \rightarrow cBd / cd$ $C \rightarrow aCd / aDd$ $D \rightarrow bDc / bc$	2 (1+1)	CO3	BL3
20.	Convert the given CFG to CNF. Consider the given grammar G: $S \rightarrow XB \mid AA$ $A \rightarrow a \mid SA$ $B \rightarrow b$ $X \rightarrow a$ and $S \rightarrow a \mid aA \mid B$ $A \rightarrow aBB \mid \epsilon$ $B \rightarrow Aa \mid b$	2 (1+1)	CO3	BL3
21.	(a). Design a PDA for accepting a language $\{anb2n \mid n \geq 1\}$. (b). Design a PDA for accepting a language $\{0^n 1^m 0^n \mid m, n \geq 1\}$. (c). Construct a PDA that accepts the language L over $\{0, 1\}$ by empty stack which accepts all the string of 0's and 1's in which a number of 0's are twice of number of 1's. (d). Design NPDA for Palindrome strips. (e). $L = \{a^m b^{m+n} c^n \mid m, n \geq 1\}$	2 (1+1)	CO4	BL4

Q. No.	Question	Marks	CO	BL
22	(a) Design a PDA for $L = \{ 0^n 1^n \mid n \geq 1 \}$. Give its formal definition, the transition rules and the sequence of instantaneous descriptions (IDs) for the input 000111. State the equivalence theorem relating to CFGs and PDAs.	2 (1+1)	CO4	BL4
23	(a) Design a Turing machine that decides $L = \{ 0^n 1^n \mid n \geq 0 \}$. Give the informal crossing-off algorithm, the transition table and the sequence of configurations for the input 0011. (b) Distinguish between Turing-recognizable and Turing-decidable languages with one example of each. Explain why multi-tape and non-deterministic Turing machines are no more powerful than the basic model and state the Church–Turing thesis.	2 (1+1)	CO3	BL3
24	(a) Prove that $A_{TM} = \{ \langle M, w \rangle \mid M \text{ is a Turing machine that accepts } w \}$ is undecidable, using the diagonalization argument, and explain why the halting problem is unsolvable. (b) Define the classes P and NP in terms of deciders and polynomial-time verifiers, define NP-completeness and state the Cook–Levin theorem. Outline the polynomial-time reduction of 3SAT to CLIQUE.	2 (1+1)	CO5	BL5
25	Construct a Turing Machine (TM) that accepts the language $L = \{ a^n b^n \mid n \geq 1 \}$. Apply your machine for the following strings aaabbb, aabbb, aabbab	2	CO5	BL5
26	Design a TM that accepts all strings over $\{a, b\}$ containing equal number of a's and b's.	2	CO5	BL5
27	Construct a TM to increment a binary number by 1 (input is a binary string).	2	CO5	BL5
28	Construct a TM that accepts the language $L = \{ ww^r \mid w \in \{0,1\}^+ \}$ where w^r is the reverse of w .	2	CO5	BL5
29	(a) Construct a TM that accepts $L = \{ a^n b^n c^n \mid n \geq 1 \}$. (b) Construct a TM that recognizes $L = \{ ww \mid w \in \{0,1\}^+ \}$.	2+2	CO5	BL5
30	(a) Differentiate between recursive and recursively enumerable languages with TM examples. (b) Explain the role of tape, head, and control unit in the functioning of a TM. (c) Discuss how a TM can simulate any computer algorithm.	(2+2+2)	CO5	BL5

ANNEXURE – I

COVER SHEET (to be completed and attached as the first page of the submission)

Name of the Student	
Registration Number	
Programme / Branch	
Semester / Section / Batch	
Course Code and Title	 / Automata and Formal Languages
Assignment Number	01
Date of Submission	
Course Coordinator	Dr. Shinnu Jangra, Department of AI & ML, MUJ
Marks Awarded (out of 20)	 Scaled Marks (out of 10):
Signature of the Evaluator	

Declaration by the Student

I declare that the work submitted in this assignment is my own. I have not copied it, in part or in full, from any classmate, book, website or generative-AI tool without acknowledgement, and I understand that any such copying will attract zero marks under the University's academic integrity policy.

Date:

Signature of the Student: